

Samarth Sethi

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EDUCATION

Johns Hopkins University

Expected May 2029

Bachelor of Science, Major in Mechanical Engineering, Minor in Systems Engineering *GPA: 3.95 / 4.0, Dean's List*
Courses: Python; Linear Algebra; Differential Equations; Design/CAD; MATLAB; Thermodynamics; Statics/Mechanics

EXPERIENCE

Weihs Lab Researcher - Johns Hopkins Extreme Materials Institute

April 2026 - Present

- Co-authored corrosion-fatigue study of biodegradable surgical stent material (Mg-Al) under 3-point bend
- Operated furnace to heat treat Mg-Al up to 450 C; executed **CNC Wire EDM** to cut 1-2 mm thick samples, optimizing **g-code** and clamping to accelerate project completion **by 25%** and **4-6x** sample production
- Adopted PhD student's job in applying precise **lacquer** to all contact points of **micrometer**-measured samples to prevent unwanted crack propagation during cyclic loading and ensure evenly-spaced grips
- Led 4 corrosion-only experiments to corroborate 3-point bending results, designing clamping apparatus in **SolidWorks** to suspend 12-15 5x5x2 mm Mg-Al samples in corrosive salt solution for three days; reduced assembly time **by 80%** and part manufacturing **by 66%**

Researcher in Johns Hopkins Multi-Hazard Resilient Structures Lab

January 2026 - Present

- Leading physics modeling of fire growth in cold-formed steel structures under **\$600,000 NSF award**
- Developed **6+ 2D FEA** thermal models of protected cold-formed steel wall assemblies exposed to experimental fires, producing **2 new** findings from parametric analysis of gypsum and gas temperature
- Presented graduate research group with validation of measured experimental temperatures using FE results

Design Build Fly Club (AIAA Competition Team) - Johns Hopkins University

September 2025 - Present

- Conducted carbon-fiber torsion analysis (**ANSYS**) to avoid **~\$250** in cost and **1-2 month** fuselage re-design
- Led manufacturing process for banner frame and deployment; designed linear actuator mechanism in **SolidWorks** to unfurl and drop the banner; prevented **4 failure modes** and optimized for success on first try
- Successful banner mission yields most score, placing us **17th out of 175** global teams (best in club history)

Product & Data Lead at Kamoky (Internship at Social App for Ethnic Products)

May 2026 - August 2026

- Built live, RAM-optimized **Python-Flask** dashboard CEO views for free, saving **\$25/month** while computing **6 new** retention metrics and category-labeled posts to benefit sales team beyond past platform
- Performed detailed **SQL** queries to discover in-app gaps between diaspora users' needs and advertised products, culminating in first-kind report shown to leaders of **U.S.-African government** consultancies
- Aided **ethnic businesses** in improving their posts' reach and viewer info (i.e. topics viewed most, location, retention by location, activity by month); developed under **MATLAB** GUI and Python backend
- Devised accurate, instant logging of clicks on app store links promoted by media influencers, including conversion to app downloads and log-in, by leveraging **Mixpanel-MongoDB** device metadata matching

LEADERSHIP

President of Futurism at JHU Podcast ([Linktree](#))

January 2026 - Present

- Manage a team of 10 students, a club with 90 members, to deliver successful recordings and guest outreach
- **100,000+** listens over **30+** countries, hosting leaders in neurology, climate, cybersecurity, biodiversity, AI

Business Development Associate for Lumiere Education

October 2025 - April 2026

- Communicated with **100+** students, clubs, faculty, and career center to promote remote work opportunities

SKILLS AND CERTIFICATES

CAD: Onshape, SolidWorks, Fusion, NX; **FEA:** ANSYS; **MFG:** EDM, 3D/Resin Printing, Laser Cutting, Drill, Band Saw
Systems Thinking: Map the System Program Certificate: **University of Oxford** *Received August 21, 2026*